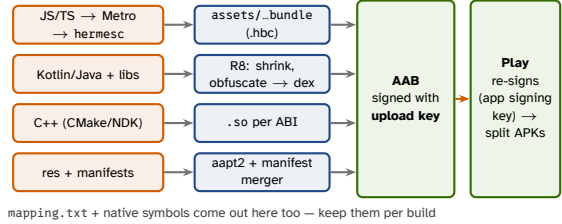


In one line: android/ is an ordinary **Gradle** project embedding a JS bundle: **AGP** compiles Kotlin/Java (R8 shrinks it), the **NDK** builds C++ per ABI, the **manifest merger** combines every library's manifest, you sign with an **upload key** and Play re-signs. Most Android “RN bugs” are manifest, permission, SDK-level or R8 problems.

android/ — who owns what

settings.gradle	modules, RN Gradle plugin, autolinking
build.gradle	minSdk 24 · compileSdk 37 · targetSdk 36, Kotlin 2.2 (0.87)
gradle.properties	ABIs, hermesEnabled, edgeToEdgeEnabled, JVM args, AGP 9 opt-outs
gradle/wrapper/	Gradle version — ./gradlew pins it
local.properties	sdk.dir — machine-local, never commit
app/build.gradle	react {}, applicationId, versionCode, signing, buildTypes
app/proguard-rules.pro	R8 keep rules (reflection, JNI)
app/src/main/AndroidManifest.xml	permissions, activities, intent filters
app/src/main/java/com/example/app/	MainActivity.kt, MainApplication.kt
app/src/main/res/	icons (mipmap), splash theme, strings.xml

./gradlew bundleRelease



Gradle, SDK levels, Play rules

- **Three SDK numbers:** minSdk = oldest device that installs; compileSdk = APIs you can call; targetSdk = which new behaviours you opt into.
- **Play target API:** since **31 Aug 2026** new apps and updates must target **API 36** (Wear/Automotive 35, TV/XR 34; extension to 1 Nov 2026 on request); existing apps below 35 stop reaching new users on newer Android.
- **16 KB pages:** targets Android 15+ must support 16 KB-page 64-bit devices (Android guide: updates blocked from **1 Feb 2027**). AGP 8.5.1+/NDK r28+ align by default; RN is compliant since 0.77 — check third-party .so in APK Analyzer.
- **Toolchain:** JDK **17**; AGP comes via RN's Gradle plugin; **0.87** adds **AGP 9** (android.builtInKotlin=false, android.newDsl=false). Since 0.82 newArchEnabled=false is *ignored*.

Manifest

- **Merged:** every library adds its manifest — read Android Studio's *Merged Manifest* tab; drop an unwanted permission with tools:node="remove".
- **android:exported** must be explicit on any component with an intent filter (Android 12+), else the app *won't install*. Launcher activity: true.
- Template MainActivity: singleTask (deep links reuse it), adjustResize, configChanges (rotation keeps the React root). App Links: autoVerify + /.well-known/assetlinks.json.

Example — manifest + permission

```
<uses-permission android:name="android.permission.CAMERA"/>
<activity android:name=".MainActivity" android:exported="true"
  android:launchMode="singleTask"
  android:windowSoftInputMode="adjustResize">
  <intent-filter android:autoVerify="true">
    <action android:name="android.intent.action.VIEW"/>
    <category android:name="android.intent.category.DEFAULT"/>
    <category android:name="android.intent.category.BROWSABLE"/>
    <data android:scheme="https" android:host="example.com"/>
  </intent-filter></activity>
<!-- JS: await PermissionsAndroid.request(PERMISSIONS.CAMERA)
  'granted' | 'denied' | 'never_ask_again' -> openSettings -->
```

Runtime permissions

- **Dangerous** permissions (Android 6+) need manifest *and* a runtime request: check → rationale (shouldShowRequestPermissionRationale) → request. POST_NOTIFICATIONS is runtime since Android 13.

- Android 11+: **Deny twice = “don’t ask again”** — the dialog never shows, send users to Settings. “Only this time” for location/camera/mic; unused apps get permissions **auto-reset**. Re-check on every resume.

Back, edge-to-edge, R8, signing

- **BackHandler:** listeners run **LIFO**; return true consumes; none → default back. An open Modal blocks BackHandler — use onRequestClose.
- **Target 36:** predictive back on by default, onBackPressed not called, KEYCODE_BACK not dispatched (BackHandler still works for most cases; opt-out enableOnBackPressedCallback="false"); **edge-to-edge** can't be opted out — use safe-area insets.
- **R8** keeps only what it sees: reflection, JNI, JSON-by-field-name need -keep class ... / @Keep.
- **Signing:** keytool -genkeypair ... -validity 10000 → upload key; passwords in ~/.gradle/gradle.properties, not git. **Play App Signing** holds the real key, so a lost *upload* key can be reset.

Build errors → fix

- **SDK location not found** → ANDROID_HOME / local.properties. **Java version errors** → JDK 17 (JAVA_HOME).
- **Duplicate class ... in modules** → two versions of one library: align/exclude. **Incompatible Kotlin metadata** → align Kotlin versions.
- **INSTALL_FAILED_UPDATE_INCOMPATIBLE** → different signing key: uninstall. **Unable to load script** → adb reverse tcp:8081 tcp:8081.
- **Release-only ClassNotFoundException** → keep rules. Stale native build → ./gradlew clean, delete app/.cxx.

Traps

- ! The template's release uses the **debug keystore** — replace it before shipping.
- ! Bumping targetSdk “just for Play” — it switches on new behaviour (back, edge-to-edge); test it like a feature.

Remember

min = who installs · compile = what you call · target = how you behave. You sign uploads; Play signs installs.